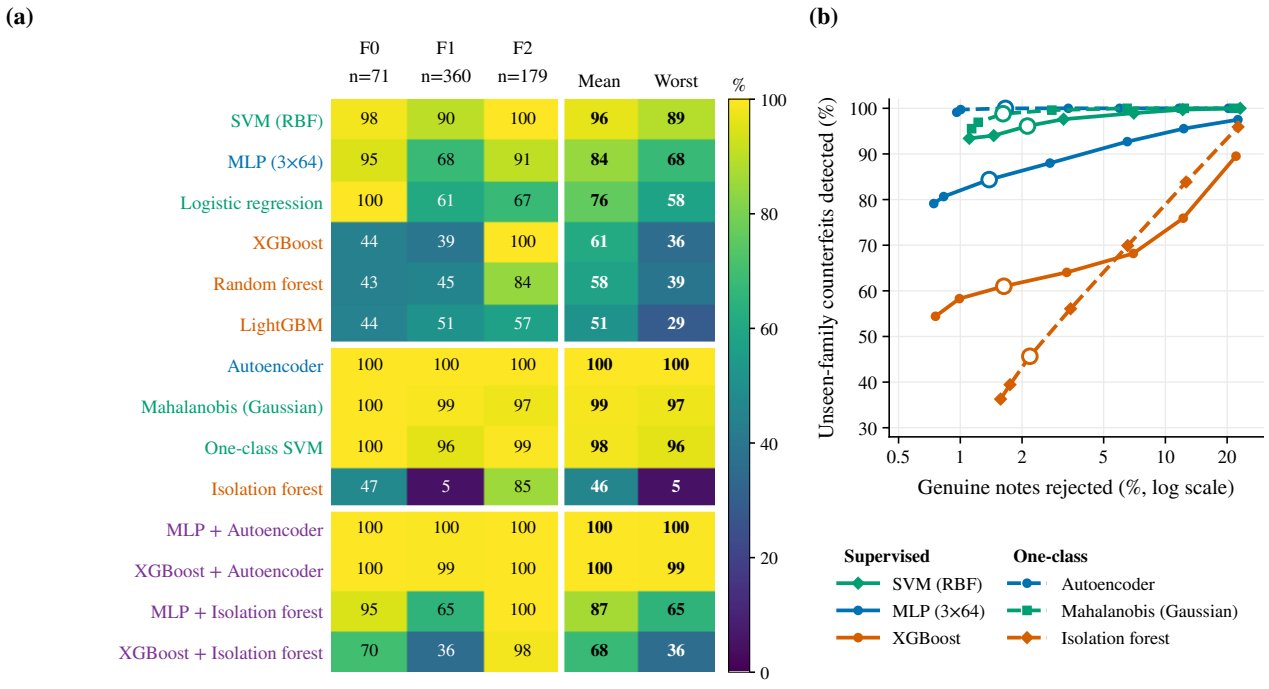
(a)
F0
n=71
F1
n=360
F2
n=179
Mean
Worst
%
100
80
60
40
20
0
SVM (RBF) 98 90 100 96 89
MLP (3×64) 95 68 91 84 68
Logistic regression 100 61 67 76 58
XGBoost 44 39 100 61 36
Random forest 43 45 84 58 39
LightGBM 44 51 57 51 29
Autoencoder 100 100 100 100 100
Mahalanobis (Gaussian) 100 99 97 99 97
One-class SVM 100 96 99 98 96
Isolation forest 47 5 85 46 5
MLP + Autoencoder 100 100 100 100 100
XGBoost + Autoencoder 100 99 100 100 99
MLP + Isolation forest 95 65 100 87 65
XGBoost + Isolation forest 70 36 98 68 36
(b)
Unseen-family counterfeits detected (%)
100
90
80
70
60
50
40
30
0.5
1
2
5
10
20
Genuine notes rejected (%, log scale)
Supervised
SVM (RBF)
MLP (3×64)
XGBoost
One-class
Autoencoder
Mahalanobis (Gaussian)
Isolation forest